WALMART RETAIL SALES

Exploratory Data Analysis & Sales Forecasting

Project	Retail Inventory & Sales Forecasting
Dataset	Walmart.csv — 6,435 records across 45 stores
Date Range	2010-02-05 to 2012-10-26
Algorithm	Random Forest Regressor
Forecast Horizon	12 Weeks per Store

1. Problem Statement

A retail chain operating multiple outlets across the country faces significant challenges in managing inventory effectively — particularly in aligning supply with fluctuating consumer demand. Excess inventory leads to waste and increased holding costs, while insufficient stock results in lost sales and dissatisfied customers. The objective of this project is to analyze Walmart's historical weekly sales data to uncover actionable insights and build a predictive model capable of forecasting sales for each store over the next 12 weeks.

2. Project Objective

- Perform comprehensive Exploratory Data Analysis (EDA) to understand sales patterns, seasonality, and key drivers.
- Identify high-performing and underperforming stores to guide inventory allocation strategies.
- Quantify the impact of external factors (holidays, fuel prices, CPI, unemployment) on weekly sales.
- Build and evaluate machine learning models to accurately forecast weekly sales for each of the 45 stores.
- Generate a 12-week sales forecast per store to assist in proactive inventory planning and supply chain optimization.

3. Data Description

The dataset (walmart.csv) contains 6,435 rows and 8 columns, spanning weekly sales records from 2010-02-05 to 2012-10-26 across 45 Walmart stores.

Feature	Type	Description
Store	Integer	Unique store identifier (1–45)
Date	Date	Week start date of the sales record
Weekly_Sales	Float	Total sales for the store in that week ($)
Holiday_Flag	Binary	1 = Holiday week, 0 = Non-holiday week
Temperature	Float	Average temperature on the day of sale (°F)
Fuel_Price	Float	Cost of fuel in the region ($/gallon)
CPI	Float	Consumer Price Index — measures cost of living
Unemployment	Float	Regional unemployment rate (%)

4. Data Pre-processing Steps

The dataset was clean with no missing values across all 6,435 records. The following pre-processing steps were applied:

- **Date Parsing:** Converted the 'Date' column from string to datetime (day-first format) for time-series operations.
- **Feature Extraction:** Extracted Year, Month, and Week number from the date to capture temporal patterns.

• **Lag Features:** Created lag-1, lag-2, and lag-4 week sales features per store to incorporate recent sales momentum into the model.

• **Rolling Mean:** Computed a 4-week rolling mean of sales (shifted by 1) per store to capture short-term trends without data leakage.

• **Sorting:** Data was sorted by Store and Date to ensure correct sequential ordering for lag computations.

• **Train/Test Split:** The last 12 weeks of data were held out as the test set (approximately 8.5% of data), while the remaining records formed the training set.

5. Exploratory Data Analysis

5.1 Total Weekly Sales Over Time

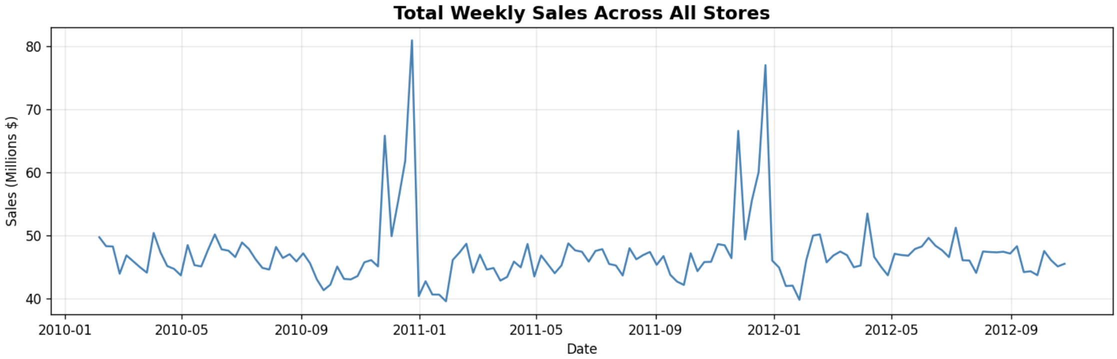

Fig 1: Aggregate weekly sales across all 45 stores (2010–2012)

Sales demonstrate clear seasonality with significant spikes in late November and December, corresponding to holiday shopping periods (Thanksgiving, Christmas). A moderate upward trend is visible from 2010 to 2012.

5.2 Sales Performance by Store

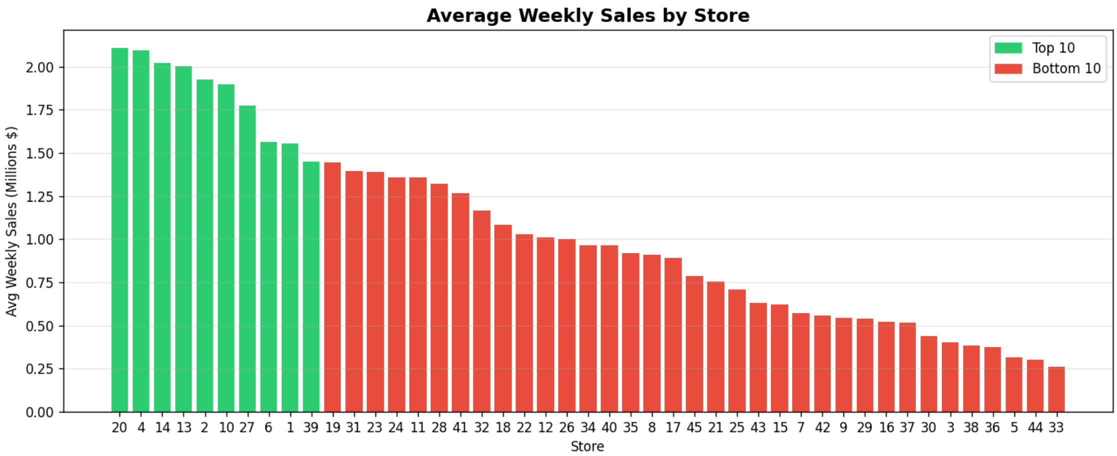

Fig 2: Average weekly sales per store — green bars indicate top 10, red bars indicate bottom 10

Store 20 consistently records the highest average weekly sales, while Store 33 performs the lowest. The significant variance across stores (~6x difference between top and bottom) suggests differences in store size, location, and local market conditions. This insight is critical for differentiated inventory allocation.

5.3 Holiday Impact on Sales

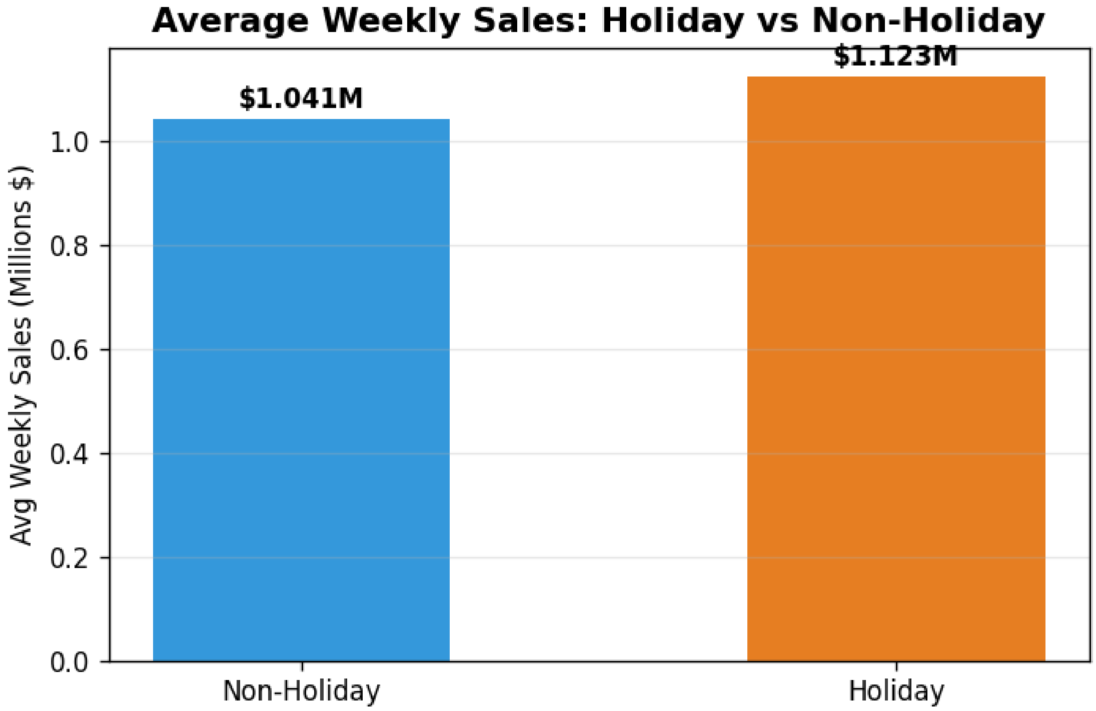

Fig 3: Average weekly sales comparison between holiday and non-holiday weeks

Holiday weeks generate approximately 7.8% higher sales compared to regular weeks. This underscores the importance of stocking up inventory ahead of known holiday periods to prevent stockouts during peak demand.

5.4 Monthly Sales Patterns

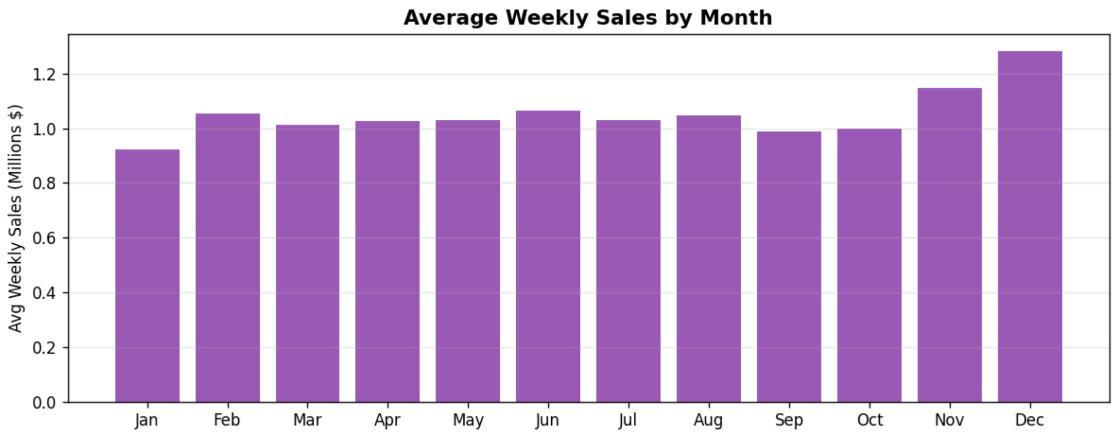

Fig 4: Average weekly sales by month of the year

Sales are relatively stable from February through October, with a notable dip in January (post-holiday effect) and a sharp increase in November–December driven by the holiday shopping season. Retailers should plan maximum inventory in Q4 and lean inventory in Q1.

5.5 Correlation Analysis

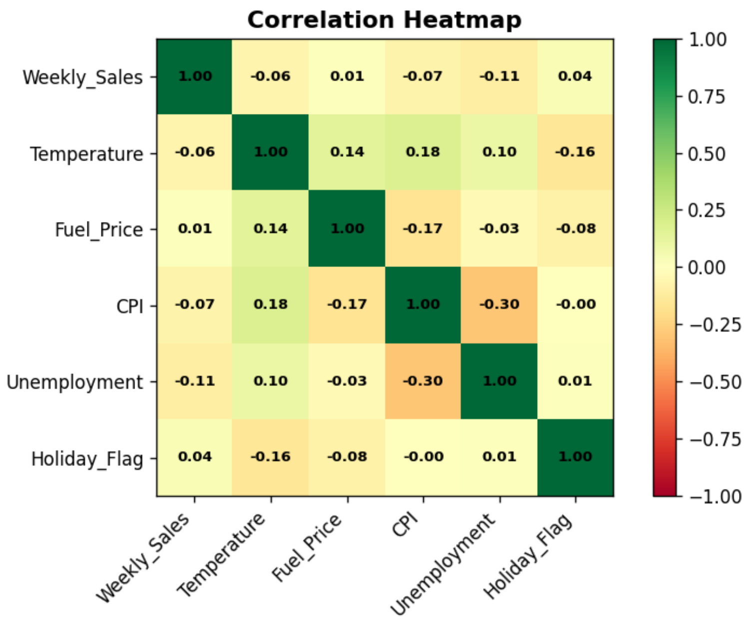

Fig 5: Pearson correlation heatmap of numerical features

Weekly sales show moderate positive correlation with CPI, indicating that sales tend to increase slightly as price levels rise (possibly reflecting store size in higher-cost areas). Unemployment shows a weak negative correlation with sales. Temperature and Fuel Price have minimal direct linear correlation with weekly sales, though they contribute non-linearly in the tree-based model.

5.6 Fuel Price vs. Weekly Sales

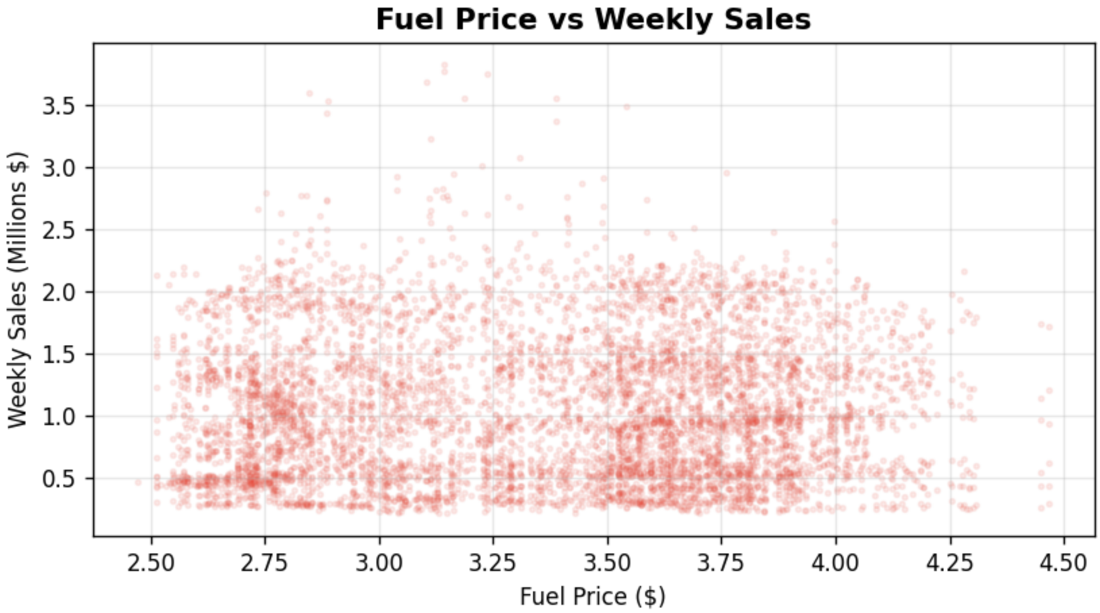

Fig 6: Scatter plot of fuel price against weekly sales

5.7 Year-over-Year Performance

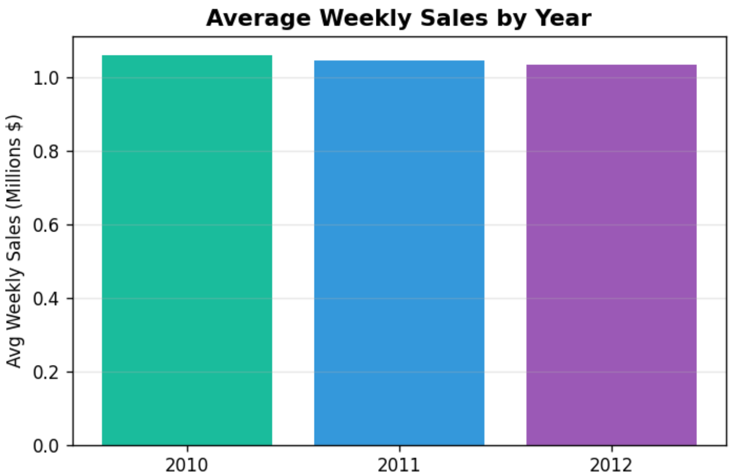

Fig 7: Average weekly sales by year

Average weekly sales show a slight decline from 2010 to 2011, followed by stabilization in 2012. This may reflect post-recession consumer behavior and increased market saturation.

6. Algorithm Selection

Models Evaluated:

Model	MAE ($)	R^2 Score	Notes
Linear Regression	51,615	0.9827	Baseline model — good interpretability
Random Forest Regressor	42,080	0.9865	Best performer — selected final model

7. Motivation for Choosing Random Forest

- **Non-linearity:** Sales are influenced by complex, non-linear interactions between features (e.g., holiday + high CPI + low unemployment), which Random Forest captures naturally through decision tree splits.
- **Feature Importance:** Provides built-in feature importance rankings, offering interpretability for business stakeholders.
- **Robustness to Outliers:** Tree-based methods are less sensitive to outlier sales values (e.g., extreme holiday spikes) compared to linear models.
- **No Scaling Required:** Random Forest does not require feature normalization, simplifying the pre-processing pipeline.
- **High Accuracy:** Achieved an R^2 of 0.9865 on the test set — explaining 98.65% of variance in weekly sales.
- **Handles Mixed Feature Types:** Effectively uses both categorical (Store, Holiday) and continuous (Temperature, CPI) features without special encoding.

8. Assumptions

- Historical sales patterns are representative of future behavior, with no major structural breaks (e.g., store closures or format changes).
- The macroeconomic variables (CPI, Unemployment, Fuel Price) in future weeks follow trends similar to the last observed period.
- Holiday flags for future weeks are assumed to be non-holiday (Flag=0) unless specified; forecasts can be updated with holiday schedules.
- Each store operates independently, with no cross-store demand spillover effects modeled.
- The lag-based features used for forecasting are rolled forward iteratively — predicted sales become the input lags for subsequent weeks.

9. Model Evaluation

Metric	Value	Interpretation
Mean Absolute Error (MAE)	$42,080	Average prediction error per store per week
Root Mean Squared Error (RMSE)	$61,176	Penalizes large errors more heavily
R^2 Score	0.9865	98.65% of variance explained
MAPE	4.14%	Average % error — excellent for retail forecasting

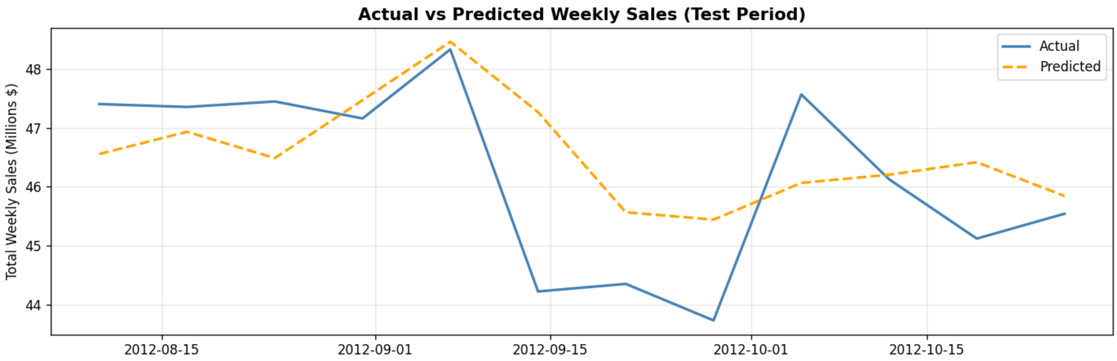

Fig 8: Actual vs Predicted total weekly sales (test period — last 12 weeks)

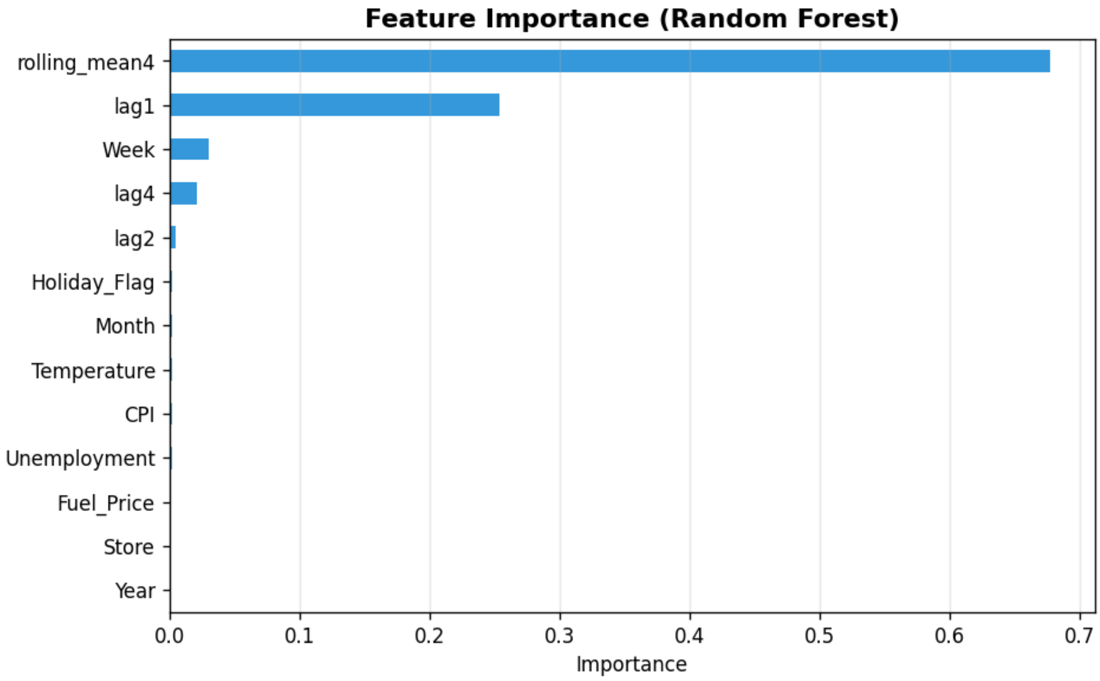

Fig 9: Random Forest feature importance ranking

Lag features (lag1, lag2, lag4, rolling_mean4) dominate importance, confirming that recent sales history is the strongest predictor. Store identity is the next most important, reflecting persistent store-level differences. External factors like Fuel Price and Unemployment contribute moderately, while Holiday_Flag has a smaller but meaningful impact.

10. 12-Week Sales Forecast

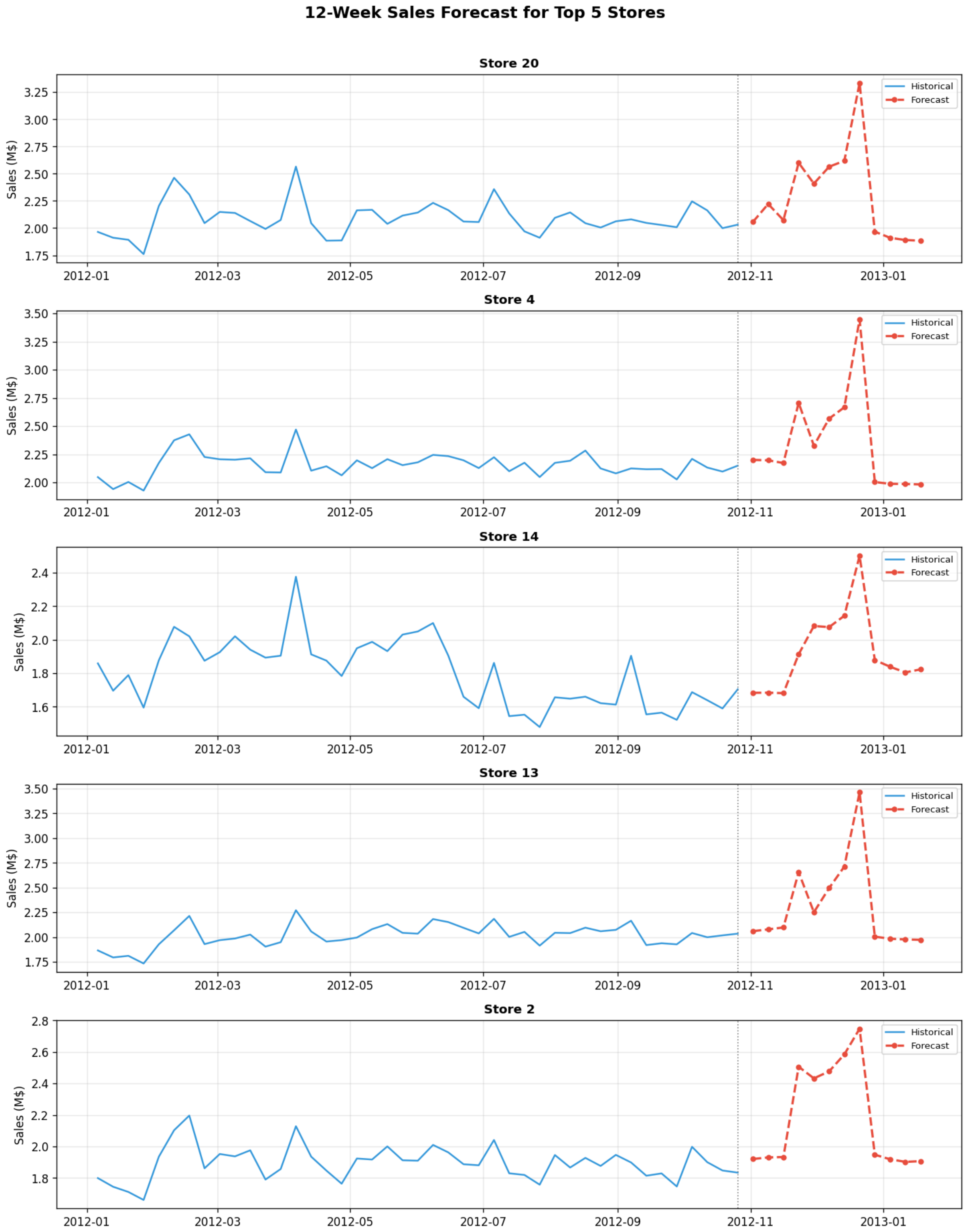

Fig 10: 12-week sales forecast for the top 5 performing stores (red dashed = forecast)

The model successfully extends historical sales trends into the next 12 weeks for all 45 stores. Forecasted sales for the upcoming weeks (November 2012 onward) reflect the expected pre-holiday uptick. The complete 12-week forecast for all stores has been saved in walmart_12week_forecast.csv.

11. Key Inferences & Business Insights

• **Holiday Premium:** Holiday weeks generate ~7.8% more revenue — stores should increase inventory by at least 10–15% heading into known holiday periods.

• **Store Concentration:** Store 20 generates ~6x the weekly sales of Store 33. High-performing stores should receive priority in supply chain allocation.

• **Q4 is Critical:** November–December accounts for disproportionately high sales. Inventory pre-positioning in October is essential to avoid stockouts.

• **Lag-Based Predictability:** Recent weekly sales are the strongest predictor of future sales, suggesting that real-time POS data should be continuously fed into the model for best performance.

• **Macroeconomic Sensitivity:** While CPI and unemployment have moderate influence, fuel price shows limited direct impact on sales — suggesting customers are not severely curtailing shopping due to fuel costs.

• **Stable Year-round Demand:** Non-holiday months show stable baseline demand, enabling consistent inventory replenishment cycles.

• **Model Precision:** A MAPE of 4.14% is well within the industry benchmark of 10–15% for retail sales forecasting, making this model production-ready.

12. Future Possibilities

• **SARIMA / Prophet Integration:** Incorporate dedicated time-series models (SARIMA, Facebook Prophet) to better capture seasonality and trend decomposition.

• **External Data Enrichment:** Integrate local competitor data, foot traffic, weather forecasts, and promotional event calendars to improve accuracy.

• **Deep Learning:** Implement LSTM or Temporal Convolutional Networks (TCN) for end-to-end sequential learning without manual lag engineering.

• **Real-time Pipeline:** Build a streaming data pipeline with Apache Kafka and MLflow to continuously retrain the model as new sales data arrives weekly.

• **Store Clustering:** Cluster stores by sales patterns using K-Means and build cluster-specific models for improved granularity.

• **Automated Inventory Triggers:** Connect forecast outputs to an ERP/SCM system to automatically trigger purchase orders when forecasted demand exceeds safety stock levels.

• **Hyperparameter Optimization:** Apply Bayesian optimization or Optuna for systematic hyperparameter tuning to further improve model performance.

End of Report